The Cardboard Box Economy Is Breaking Down

Podcast: Wall street journal

AI Executive Summary

Core Thesis

While per capita cardboard consumption has declined since its peak in 1999 due to packaging optimization and recycling efforts, the U.S. pulpwood market is facing a crisis that contradicts these trends. The collapse of domestic manufacturing facilities and competition from cheaper foreign eucalyptus pulp have driven wood fiber prices near zero, turning what was historically a profitable byproduct for timber farmers into an economic liability. This has triggered an existential threat to family sawmill operations in the American South, forcing landowners to reconsider whether growing trees should continue to be their primary use.

Key Arguments & Data

- **Demand vs. Price Paradox:** Per capita cardboard usage peaked in 1999 and is currently lower due to smaller boxes, more plastic mailers, and increased recycling; however, pulpwood value collapsed alongside demand reductions.
- **Economic Collapse:** Pulpwood prices dropped from roughly $15–$16 per ton to approximately $0 per ton, creating a situation where farmers are being charged to have trees hauled away for disposal.
- **Competition from Abroad:** A flood of cheap eucalyptus pulp from South America has flooded the market, undercutting domestic tree prices significantly due to lower labor costs and faster growth rates.
- **Manufacturing Contraction:** Major international paper mills have permanently closed facilities in key regions, such as Savannah and Riceboro, Georgia (owned by International Paper).
- **Seedling Market Data:** Sales of loblolly seedlings fell by 20% over the last two fiscal years.
- **Farmers' Adaptation:** Southern tree farmers reduced planting density from 650 trees per acre to as low as 450 trees per acre to minimize pulpwood inventory that yields no return.

- **Generational Impact:** Some farms are transitioning to blueberry agriculture, solar leasing, or small-scale ranching, whereas others remain hopeful but face a 130-year generational transition risk.

Actionable Takeaways

- **For Investors and Landowners:** Assess the risk profile of timber assets in the Southern U.S.; consider diversifying land use away from traditional pulpwood-focused forestry if local manufacturing infrastructure continues to decline.
- **For Policymakers:** Monitor for potential subsidies or market interventions regarding sustainable biomass markets (e.g., sustainable aviation fuel, cellulose textiles) which offer alternative revenue streams for timber farmers.
- **For Industry Stakeholders:** Evaluate the long-term viability of investing in pulpwood monocultures against emerging alternatives like OSB manufacturing or carbon credit markets as pulp demand continues to shift away from traditional newsprint and shipping boxes.
- **Consumer Awareness:** Understand that changes in paperboard production could impact product availability and cost, particularly in sectors reliant on specific cellulose fibers for sustainable packaging or textiles.

Full Transcript

Okay, I'm Ryan December. I am a reporter for the Wall Street Journal. That's a great name you have there, Ryan. Yeah, it is. Good one for you too. Two Rhyans in two different studios here to talk about one cardboard box. You're holding one in your hand right now. Yeah, I grabbed one because they're everywhere these days. The technical term is a corrugated packaging. Cardboard is actually more like your paper cup, but we don't have to quibble about that. We won't. Cardboard, container board, you know what I'm talking about. The brown shipping box is maybe the least exotic object in the world. The stuff is everywhere, crammed in recycling bins on doorsteps, stacked inside e-commerce warehouses. Yet if you thought we were living in the era of peak cardboard, you'd actually be wrong.

So, you probably wouldn't guess it, but we're actually on a per capita basis. America's using a lot less cardboard than we used to. Really? Yeah. When you break it down of boxes per person, we peaked in 1999, which is sort of hard to believe. I'm surprised that there's less container board being used now in an era of like online shopping than there was 20 years ago.

Think about like when you used to get a delivery from Amazon and you'd open it, it'd be like a box inside another box with another little box with the small thing you ordered. Technology and like a lot of sort of work on their end has gotten rid of that, right? Not only did the box inside the box inside the box start disappearing, but boxes themselves have also gotten smaller and lighter. More stuff started showing up in plastic mailers and more boxes are getting made from old recycled ones. So, I mean like overall like this sounds good. There's less waste. There's more recycling. But there obviously must be a downside here to somebody otherwise we wouldn't be talking. Yes. The downside is if you were selling trees. Specifically pulpwood. That's a type of wood used to make cardboard. And recently in the pulpwood market like a poorly taped box, the bottom fell out.

It varies highly, you know, by market, but some markets you might have gotten 16 a ton, 15 a ton, and now you might get zero. Zero dollars a ton?

Yeah. So basically you're saying this stuff is worthless. Yes. What does this mean for the tree growers, the people whose livelihoods depend on selling these trees? It's an existential crisis. Welcome to The Journal, our show about money, business, and power. I'm Ryan Knudson. It's Thursday, September 3rd. Coming up on the show, little trees, big problems. In many parts of the American South, a forest isn't just a place where towering pine trees grow wild. Trees are a crop. Ryan December has a way of visualizing this. Go to your Google Earth or your online map, set it into satellite mode, zero in on the south, find something that's green.

not like a wildlife refuge or a national park, but any other stretch that's green. And as you zoom in, more likely than not, you're going to find that the trees are all planted in neat rows. Neat rows of carefully planted loblolly pines, tree farms. In the timber industry, the big money is in lumber. That's a processed wood that comes from mature 20 or 30-year-old trunks that eventually turn into two-by-fours telephone poles. And like any crop, there's a tried and true way to get the best harvest. They say plant them thick and cut them quick. The goal is to get tall pints with few branches. So you plant the trees really close together and that forces them to sort of compete and race up to the sky, to the sun. The idea is that the trees chasing sunlight will shoot straight up and shed their lower branches in the process, which means fewer nuts in the wood and taller trunks. The problem is that eventually the trees get overcrowded and tree farmers have to do some thinning. After a while, the soil only supports so many mature trees. So you have to go in and kind of select the ones that are the worst specimens or misshapen trees or maybe just every other one or whatever your plan is. And you have to take those out to make room for the best specimens to keep growing into lumber. So if you were to go to one of these pine plantations and look around you might see trees that are pretty unremarkable maybe they're crooked or you know otherwise not the best specimens and you can't really envision them becoming a big utility pole but they're very important to the whole system. Important to the system because those dinky crooked trees can still be sold as pulpwood and turned into paper and cardboard boxes.

And they basically make it into a mashed, almost like an oatmeal and that becomes pulp and then they roll it out in flat sheets and basically dry it and it becomes paper. In other words, if the farmer's main goal was lumber,

pulpwood was basically a profitable byproduct. In a business where farmers had to wait decades for big trees to mature, pulpwood provided the first financial return and a steady trickle of income. And then you let the trees remain and grow for another 25, 30 years. and they become lumber and that's your big payday. For almost a century, that was how it worked. But now, as we mentioned earlier, the system is breaking down. Americans are using less paper. Cardboard demand is falling. And tree farmers are getting hit from another direction. Cheap foreign competition. A flood of cheap eucalyptus pulp from South America. The eucalyptus tree grows faster, the labor's cheaper.

And then the paper mills that buy pulpwood started shutting down. International paper is closing its facilities. Very recently, just last year, we had a big rash of pulp mill closings across the south. Georgia's kind of ground zero, the big mill in Savannah, another one in a crossroads out in the country called Riceboro. These mills started closing. Communities are wrought. After learning international paper facilities in Savannah and Riceboro will soon shut down permanently. The impact is cutting even deeper on businesses that depend on the timber industry. All of these forces have helped drive down the value of pulpwood. The price for those little trees is now so low that instead of selling them, some farmers are now paying to have them hauled away. You still need to make room so you go and take out every third tree, right? And those go to the pulp mill.

And he used to be able to make money off of that. Yeah, that used to be a big payday. Now, for some, it's a big bill. That changes the economics for a lot of southern pine farmers. Because if pulpwood no longer pays, what are you supposed to do? That's the question confronting Bill Howard. He's a fourth generation sawmill owner in Statesboro, Georgia. Yes, sure. There's a way of living down here. Bill's family has been in timber for more than 130 years. Now, he worries that his way of life is in jeopardy. But if growing trees stops making sense, what happens to the farmers and the forests that they depend on? That's next. Not long ago, my colleague Ryan December went down to Georgia.

He wanted to see what the collapse of the pulpwood market looks like for people growing the trees. That's where he met Bill. Bill Howard, Bill

Howard's family is very indicative of this whole crisis. Yeah, Bill Howard, station for Georgia, fourth generation, Saul Miller. As I say, we turn round things into square things. Bill can't imagine doing anything else. The old saying is he gets saltless in your veins. He can't ever get out of it. His great grandfather turned up around Statesboro, Georgia in 1898 with seven brothers. In 1898, my great-granddad moved into Statesboro. Had a portable salt mill. And they showed up and started sawing and eventually the next generation built a permanent mill in town. And then in 45, we moved for it right now. And they've expanded into sort of like a contractor yard to sort of build or supply business.

All of a sudden, they've got 300-some employees. For years, the Howard's could count on the first thinning of the forest as a payday. Now it's a burden. The mills that make newsprint and boxes are buying less and less of Bill's pulpwood. And then those paper mills started closing. When you saw that happening, Bill started worrying about what it meant for his business, both as a farmer and the owner of a local sawmill. Well, we still have paper mills in our area. What keeps me up at night is that we lose another one. That fear is already changing the way Bill is planting his forests. So we used to plant 650 trees per acre. We've gone to 550 and then here recently we've gone to 450 just trying to avoid having more pup wood to take out. Across the south, farmers may already be planting fewer trees.

One of the region's biggest suppliers of pine seedlings reported that over its last two fiscal years, sales of loblolly seedlings have fallen 20%. So it's bad for him as a tree grower, and I'm guessing it's also not great for a sawmill business. You know, when I talked to him, he said, look, if people don't plant trees, I won't have anything to cut. Polkwood is the first step. If we don't have a manufacturing base to sell our trees into, we won't have a forest. Some farmers are considering a much bigger change, not planting trees at all. You know, they have alternatives. They could turn it into ag land. They could turn it into solar farms. They could develop it. They're going to do something with that land for best use. We want it to be growing trees. One grower in the region told Ryan that instead of replanting pine trees on part of his land, he planted blueberries instead. Others are leasing their land for solar.

or considering selling off parcels as little ranches. Is there a chance there could be a new market for the pulpwood? Like they use it for something else? Yes. There's a lot of things happening and a lot of efforts. Like OSB, it's called Oriented Strandboard. It's basically like the floor or the roof of your house, manufactured panels. And then there's, like, more sort of interesting new things people are trying to develop, like, sustainable aviation fuel, jet fuel made from cellulose. There's fabrics. If you go through your shirts, if you have a silky one, it might be made out of something called lyosel, which is made from cellulose, or modal is basically a member of the rayon family, which is made from pulp.

you know, smokeless gunpowder for the military has made it a plant down in Georgia out of wood pulp. So there's all sorts of new things. If you're a timber farmer, though, these are a lot of really interesting ideas that, you know, it's hard to say at this point if they're going to pay off, but you need to make a decision you're going to be beholden to for decades. Exactly. And that's sort of the conundrum. It's like, do you bet on that or something like that happening? Or do you do something else? Emotionally, sort of on a human level for these farmers, like, what do you think? is going through their mind is they're trying to make a decision about whether or not to plant a tree or even stay in this business at all. I mean I was thinking about that and like you know I live in the city so like my assets are in stocks and bonds. I have zero emotional attachment to those. I don't wake up and go look at them out my window. I don't take a walk through them. I don't hear the birds chirping in them. I don't have my family over for like you know a turkey hunt or whatever.

And that's what these people do. We hunt and fish and camp out. We manage it to grow salt timber, but along the ways, you know, you see it at every age. So there's like an aesthetic attachment to the land and to the trees and the forest itself. It is. It is beautiful. I think that makes it a lot harder than if, say, you or I had some need for money. And we say, oh, well, we'll just sell those bonds or sell those stocks. But if you take my thousand acres of pine trees, that's gonna materially probably change my life and how I live it. Bill isn't ready to pull the plug on the family business yet. He now has two sons that have joined the company, a fifth generation of tree farmers. And they have kids too, meaning there's already a sixth generation running

around the family timberlands. Bill helps they'll join the family business too when they grow up. And one day harvest the trees he planted this year.

He, like a lot of other people down there, these are very hardworking people. They have a lot of faith. They tend to be optimistic. Right now, it feels like the sky's falling a little bit, but in the end, I think we'll prevail. And so he, like a lot of other people I talk to, we're kind of like, something's gonna turn around here for us if we can just kind of wait it out. We've been in this going on 130 years sawmill and so it. It's not going to end on my watch. We're going to figure it out and keep going. That's all for today. Thursday, September 3rd. The journal is a co-production of Spotify and The Wall Street Journal. If you like our show, buy The Wall Street Journal newspaper made from pulpwood and follow us on Spotify around every weekday afternoon. Thanks for listening. See you tomorrow.